

PUDINO, Russian Federation

WMO: 293130

Lat: 57.5627N

Lon: 79.4289E

Elev: 313

StdP: 14.53

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.8	-32.6	-43.7	0.4	-38.4	-37.7	0.6	-32.2	17.2	8.2	15.7	12.4	1.6	30	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	84.0	65.6	80.9	64.4	77.8	63.3	69.0	78.3	67.1	76.3	65.3	74.3	6.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	96.4	73.6	63.7	89.4	71.2	61.9	83.7	69.5	33.4	78.3	31.9	76.3	30.5	74.3	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	14.8	13.3	DB	-44.2	89.4	6.6	3.2	-48.9	91.7	-52.7	93.6	-56.4	95.4	-61.2	97.7	(4)
(5)			WB	-43.7	72.5	5.5	2.7	-47.7	74.4	-50.9	76.0	-54.0	77.5	-58.0	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.2	-3.8	3.7	19.9	35.0	48.4	60.4	63.9	58.2	47.4	34.3	14.4	2.3		
	DBStd	25.91	15.44	13.91	11.46	10.57	10.21	8.08	6.20	6.62	7.96	9.30	15.92	16.73		
	HDD50	7715	1667	1297	933	457	159	15	0	9	141	490	1069	1478		
	HDD65	12113	2132	1717	1398	900	521	177	97	228	530	951	1519	1943		
	CDD50	1206	0	0	0	7	111	328	430	263	63	4	0	0		
	CDD65	127	0	0	0	0	7	40	62	17	1	0	0	0		
	CDH74	1591	0	0	0	1	171	552	654	194	19	0	0	0		
	CDH80	351	0	0	0	0	47	134	140	28	2	0	0	0		
	WSAvg	6.0	5.6	6.0	7.1	7.4	7.0	5.4	4.7	4.9	5.5	6.2	6.4	6.2		
	PrecAvg	20.7	1.0	0.7	0.9	1.0	1.8	2.7	3.1	3.0	2.0	1.7	1.6	1.3		
	PrecMax	28.1	2.2	1.4	2.2	2.3	4.6	6.7	7.4	5.7	3.3	3.9	2.6	2.7		
(15) Precipitation	PrecMin	17.0	0.1	0.0	0.0	0.2	0.3	0.7	0.4	0.5	0.2	0.7	0.4	0.3		
	PrecStd	2.8	0.6	0.3	0.6	0.5	1.0	1.3	1.5	1.3	0.8	0.7	0.5	0.5		
	DB	29.7	34.2	48.6	67.4	85.8	87.2	88.2	83.7	76.5	62.6	42.2	33.3			
	MCWB	28.6	31.6	38.5	53.1	62.3	65.4	69.2	66.8	60.2	50.3	39.2	32.0			
	2% DB	26.5	30.6	42.3	61.2	78.8	84.1	83.6	79.3	71.1	56.6	39.0	31.4			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	25.3	28.6	34.2	48.2	58.6	64.5	67.0	64.6	58.2	47.9	36.9	30.2			
	5% DB	22.7	27.6	38.5	56.4	73.0	80.5	80.8	75.7	65.9	51.8	35.5	27.9			
	MCWB	21.7	25.7	32.7	45.7	55.6	63.6	65.6	63.5	55.4	45.6	33.9	26.7			
	10% DB	19.1	23.7	35.2	51.3	67.7	76.6	77.7	71.7	61.4	47.6	33.0	24.1			
	MCWB	18.1	21.7	31.2	42.4	53.7	62.5	64.6	62.3	53.4	43.0	31.8	23.0			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	28.8	31.9	39.1	54.1	63.3	70.2	72.5	69.9	62.5	52.4	40.3	32.6			
	MCDB	29.5	33.4	47.5	64.0	82.0	80.8	83.3	78.2	72.1	59.3	41.3	33.1			
	2% WB	25.5	28.9	35.5	49.8	60.2	67.7	69.9	67.3	59.3	49.3	37.0	30.1			
	MCDB	26.5	30.5	39.8	59.2	74.5	78.0	78.9	75.8	68.4	54.9	38.8	31.3			
	5% WB	21.8	25.8	33.4	46.3	57.4	65.7	67.8	65.2	56.7	46.2	34.0	26.8			
(33) Mean Daily Temperature Range	MCDB	22.8	27.5	37.3	55.1	70.4	76.5	76.8	72.4	64.9	51.0	35.1	27.8			
	10% WB	18.1	21.8	31.4	42.8	54.6	63.8	66.1	62.9	54.1	43.0	32.0	23.0			
	MCDB	19.1	23.7	35.2	51.0	66.4	73.8	74.7	70.0	60.4	47.2	33.0	24.1			
	5% DB	12.3	14.1	22.5	26.9	31.3	28.4	25.8	26.2	26.6	20.4	9.5	12.7			
	MCWBR	12.0	12.9	16.0	15.7	14.2	12.6	11.9	13.6	15.0	13.1	8.9	12.4			
(40) Clear-Sky Solar Irradiance	taub	0.279	0.312	0.366	0.397	0.399	0.407	0.443	0.412	0.366	0.346	0.307	0.279			
	taud	2.014	2.009	1.939	2.087	2.256	2.314	2.198	2.272	2.358	2.253	2.075	1.961			
	Ebn at Noon	171	220	238	255	265	264	250	250	245	210	157	128			
	Edh at Noon	19	30	43	43	39	37	41	36	28	24	18	15			
(44) All-Sky Solar Radiation	RadAvg	217	534	998	1392	1588	1807	1632	1253	839	410	184	127			
	RadStd	25	41	67	135	159	191	139	136	102	37	36	20			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page